

2. Madaurin 급수 공식

$$\textcircled{1} \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\textcircled{2} \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$\textcircled{3} \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \dots = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}$$

$$\textcircled{4} \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}$$

$$\textcircled{5} e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

2. Madaurin 급수 공식

$$\textcircled{6} (1+x)^p = 1 + px + \frac{p(p-1)}{2!} x^2 + \frac{p(p-1)(p-2)}{3!} x^3 + \dots$$

$$= \sum_{n=0}^{\infty} {}_p C_r x^n = \sum_{n=0}^{\infty} \binom{p}{r} x^n$$

$$= \sum_{n=0}^{\infty} \frac{p(p-1) \cdots (p-(n-1))}{n!} x^n$$

$$\textcircled{7} \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots = \sum_{n=0}^{\infty} x^n$$

$$\textcircled{8} \frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 + \dots = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$\textcircled{9} \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

$$\textcircled{10} -\ln(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1}$$

2. Madaurin 급수 공식

$$\textcircled{11} \sin^{-1}x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \dots$$

$$= x + \frac{1}{2} \times \frac{1}{3}x^3 + \frac{1}{2} \times \frac{3}{4} \times \frac{1}{5}x^5 + \dots$$

$$= x + \sum_{n=1}^{\infty} \frac{1 \times 3 \times 5 \times \dots \times (2n-1)}{2 \times 4 \times 6 \times \dots (2n)} \times \frac{x^{2n+1}}{(2n+1)}$$

$$= \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$$

$$\textcircled{12} \sinh^{-1}x = x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \dots$$

$$= x + \sum_{n=1}^{\infty} \frac{(-1)^n \times 1 \times 3 \times 5 \times \dots \times (2n-1)}{2 \times 4 \times 6 \times \dots (2n)} \times \frac{x^{2n+1}}{(2n+1)}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$$

$$\textcircled{13} \tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots$$

$$\textcircled{14} \tanh x = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \dots$$

$$\textcircled{15} \sec x = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots$$

$$\textcircled{16} \operatorname{sech} x = 1 - \frac{1}{2}x^2 + \frac{5}{24}x^4 - \dots$$

$$\textcircled{17} \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

$$\textcircled{18} \tanh^{-1} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \frac{x^9}{9} + \dots = \sum_{n=0}^{\infty} \frac{1}{2n+1} x^{2n+1}$$

<정리>

$$\textcircled{1} \sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

$$\textcircled{2} \sum_{n=1}^{\infty} n^2 x^n = \frac{x+x^2}{(1-x)^3}$$

$$\textcircled{3} \sum_{n=1}^{\infty} n^3 x^n = \frac{x+4x^2+x^3}{(1-x)^4}$$

$$\textcircled{4} \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \frac{(1-x)\ln(1-x)+x}{x}$$